Cong Yang

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Education

2023-2026 **Ph.D. in Computer Science** Wuhan, China
- School of Computer Science, Wuhan University
- Advisor: Lefei Zhang (zhanglefei@whu.edu.cn)
- GPA: 3.71/4.0

2020-2023 **M.S. in Geological Resources and Geological Engineering** Wuhan, China
- School of Geophysics and Geomatics, China University of Geosciences
- Advisor: Yanni Dong (dongyanni@whu.edu.cn)
- GPA: 3.62/4.0

2016-2020 **B.S. in Earth Information Science and Technology** Wuhan, China
- School of Geophysics and Geomatics, China University of Geosciences
- GPA: 3.35/4.0

Research Interests

Multimodal Learning, Image Captioning, Post-Training

Research Experience

2026–Present **Associative and Diversity-Aware Reinforcement Learning for Multimodal Reasoning**

Doctoral Researcher, Advisor — Prof. Lefei Zhang

- Created Associative CoT-9K by converting linear chains of thought into linked question–answer subproblems.
- Trained multimodal models in two stages: supervised fine-tuning on Associative CoT-9K, followed by group relative policy optimization (GRPO).
- Added a semantic-diversity regularizer based on Qwen3-Embedding-4B to discourage repetitive reasoning trajectories during GRPO.
- Evaluated Qwen2.5-VL-7B and Qwen3-VL-8B on multimodal reasoning benchmarks. Conducted ablation studies, analyzed reasoning trajectories and training costs, and prepared a first-author manuscript (M4).

2025–2026 **Reliable and Efficient Perception in Vision–Language and Long-Video Models**

Doctoral Researcher, Advisor — Prof. Lefei Zhang

- Developed Focus and Awaken, a training-free method for reducing hallucinations in vision–language models. It combines a Chebyshev attention penalty with dual-track contrastive decoding using less than 0.1% of feed-forward neurons. It reduced sentence- and instance-level CHAIR scores to 46.0 and 13.2 at 1.04× inference latency (M3).
- Developed UniEP for long-video understanding. A question-conditioned evidence plan guides frame selection and hierarchical token pruning. The method outperformed uniform 32-frame sampling by 2.65 points and retained 99.3% of full-token performance using only 10% of visual tokens (M2).

2023–2026 **Vision–Language Learning for Fine-Grained Image Captioning and Change Understanding**

Doctoral Researcher, Advisor — Prof. Lefei Zhang

- Developed BITA for image–text alignment. It combines multi-scale frequency features and contrastive learning with prefix causal language modeling (first-author, J4).
- Developed MGIMM for fine-grained captioning and KCFI for change captioning. MGIMM uses regional–global feature fusion and 61,783 curated instruction pairs. KCFI focuses on key-change perception through multi-task learning (first-author, J5, J6).
- Proposed learnable change slots to select bi-temporal visual features and align them with caption semantics using soft-Chamfer matching and InfoNCE. Achieved S^* scores of 81.59 on LEVIR-CC and 90.02 on WHU-CC (M1).

2021–2022 **Metric Learning and Domain Adaptation under Limited Annotations**

Graduate Researcher, Advisor — Prof. Yanni Dong

- Developed DMLOHM for hyperspectral classification with limited annotations. It combines discriminative metric learning with online hard-triplet mining.
- Improved classification accuracy by 0.13%–4.03% using only 85 labeled samples per class.
- Developed ADA-DDA for cross-domain classification. It uses attention-based sample weighting to balance marginal and conditional distribution alignment, improving accuracy by 5.36% over the next-best method.
- Contributed to three journal publications (J1, J2, J3) and a granted Chinese invention patent (CN114202697B).

Publication

Journal Articles

2025 J6 **Enhancing Perception of Key Changes in Remote Sensing Image Change Captioning**
Cong Yang, Zuchao Li, Hongzan Jiao, Zhi Gao, Lefei Zhang
IEEE Transactions on Image Processing
DOI: 10.1109/TIP.2025.3589096

2024 J5 **MGIMM: Multi-Granularity Instruction Multimodal Model for Attribute-Guided Remote Sensing Image Detailed Description**
Cong Yang, Zuchao Li, Lefei Zhang
IEEE Transactions on Geoscience and Remote Sensing
DOI: 10.1109/TGRS.2024.3497976

J4 **Bootstrapping Interactive Image–Text Alignment for Remote Sensing Image Captioning**
Cong Yang, Zuchao Li, Lefei Zhang
IEEE Transactions on Geoscience and Remote Sensing
DOI: 10.1109/TGRS.2024.3359316

2022 J3 **Attention-Based Dynamic Alignment and Dynamic Distribution Adaptation for Remote Sensing Cross-Domain Scene Classification**
Cong Yang, Yanni Dong, Bo Du, Liangpei Zhang
IEEE Transactions on Geoscience and Remote Sensing
DOI: 10.1109/TGRS.2022.3225589

J2 **Joint Distance Transfer Metric Learning for Remote-Sensing Image Classification**
Yanni Dong, Tianyang Liang, **Cong Yang**, Hui Luo, Yuxiang Zhang
IEEE Geoscience and Remote Sensing Letters
DOI: 10.1109/LGRS.2022.3149141

2021 J1 **Deep Metric Learning with Online Hard Mining for Hyperspectral Classification**
Yanni Dong, **Cong Yang**, Yuxiang Zhang
Remote Sensing
DOI: 10.3390/rs13071368

Preprints

2026 M4 **AD-GRPO: Associative Diversity-Aware GRPO for Multimodal Reasoning**
Cong Yang, Yuchun Miao, Kangzhe Yuan, Zuchao Li, Lefei Zhang
Under Review

2026 M3 **Focus and Awaken: Mitigating VLM Hallucinations via Chebyshev Attention and Neuron-Routed Contrastive Decoding**
Ziyang Ma, **Cong Yang**, Baoyuan Qi, Haojun Ai, Ping Wang, Zuchao Li
Under Review

2026 M2 **UniEP: Unified Evidence Planning for Efficient Long Video Understanding**
Yuxuan Rao, **Cong Yang**, Kangzhe Yuan, Zuchao Li, Lefei Zhang
Under Review

2026 M1 **Semantically Aligned Change Slots for Remote Sensing Image Change Captioning**
Yuanzhi Yao, **Cong Yang**, Liang Lv, Jun Wan, Lefei Zhang
Under Review

Service

Peer Reviewing

Journals

- *IEEE Transactions on Geoscience and Remote Sensing* (TGRS)
- *IEEE Transactions on Image Processing* (TIP)
- *IEEE Transactions on Circuits and Systems for Video Technology* (TCSVT)

Awards

2023 Outstanding Graduate of China University of Geosciences
2021 China National Scholarship

Skills

Research Expertise	Remote sensing image understanding and generation; Multimodal learning; Model post-training
Programming & Computing	Python, MATLAB, C/C++; PyTorch, Hugging Face Transformers, DeepSpeed; Slurm, distributed multi-GPU computing, Git
Languages & Writing	Mandarin Chinese (native); English (professional working proficiency; experienced in academic writing and manuscript preparation)